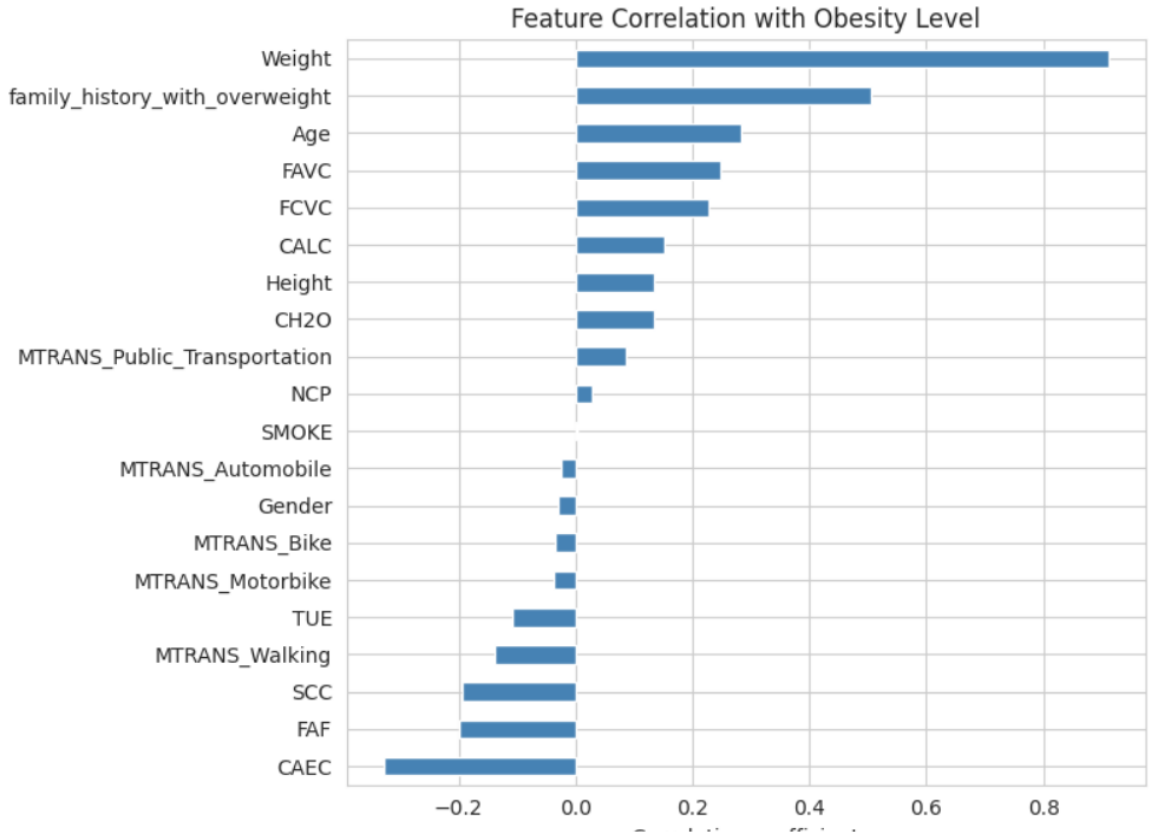
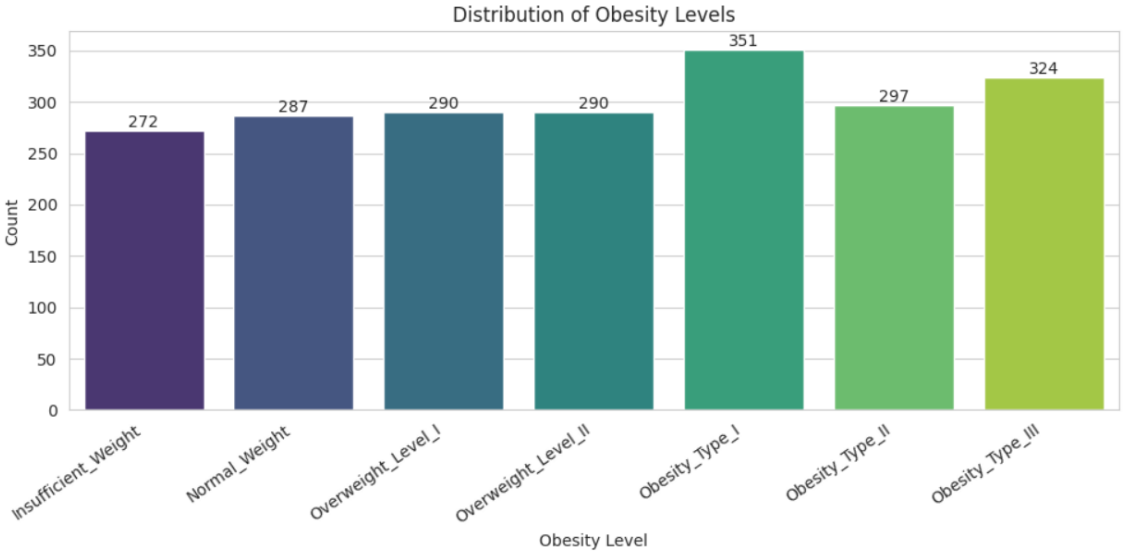
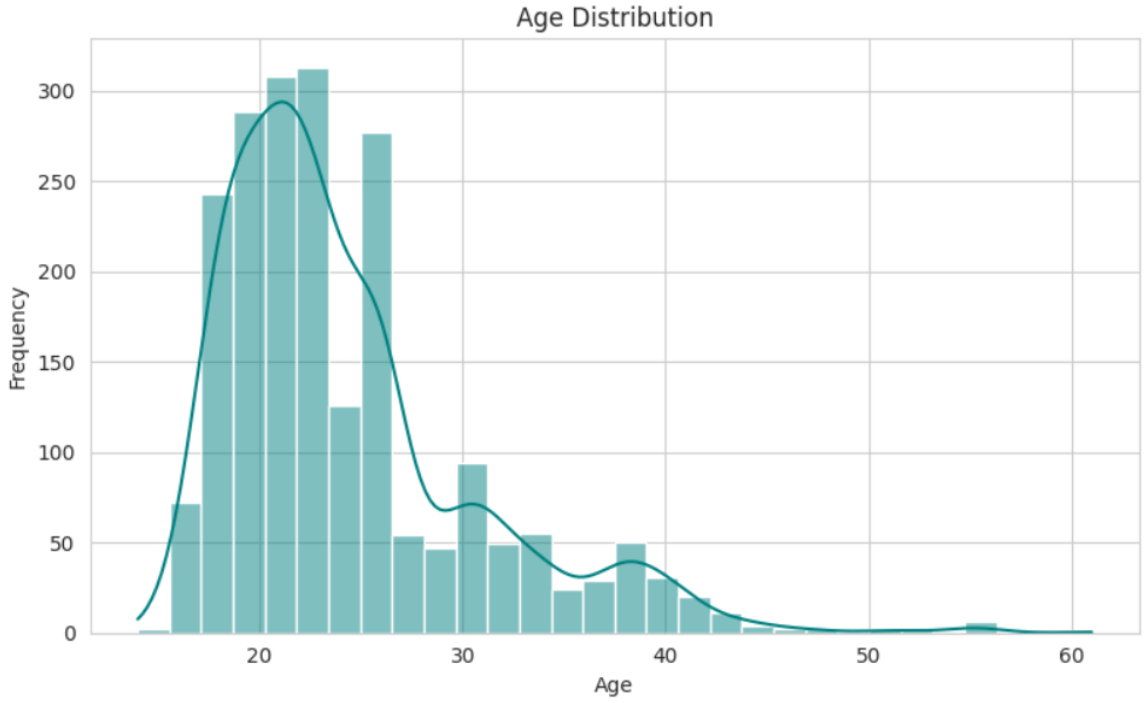
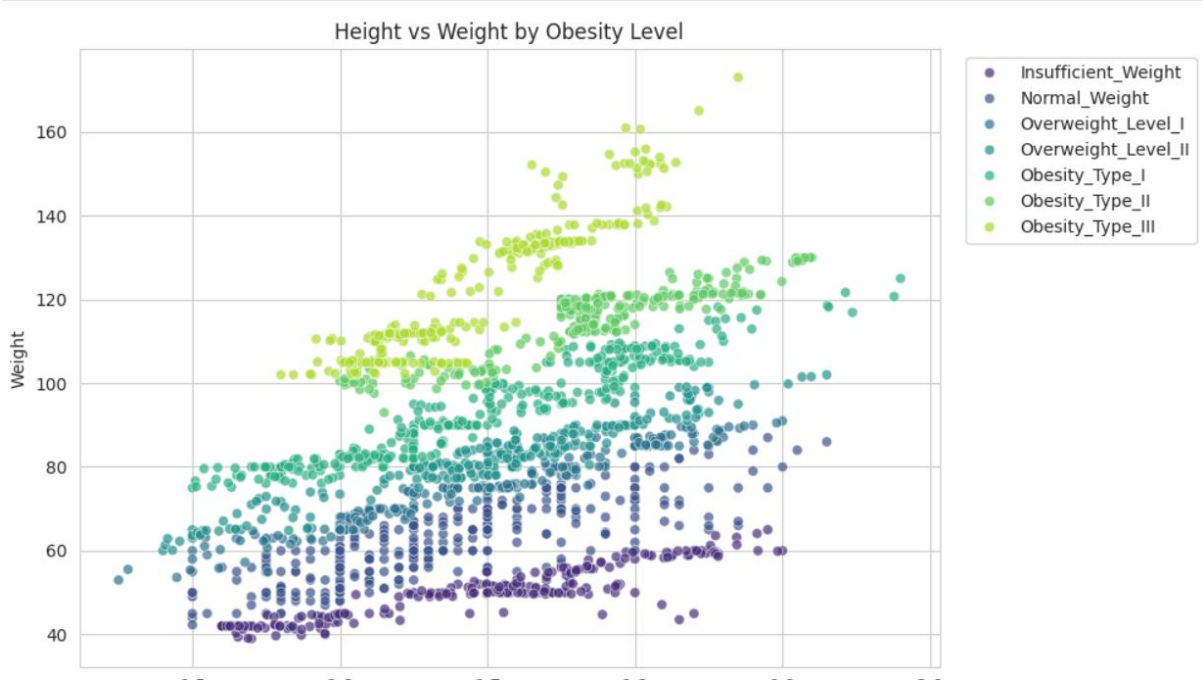
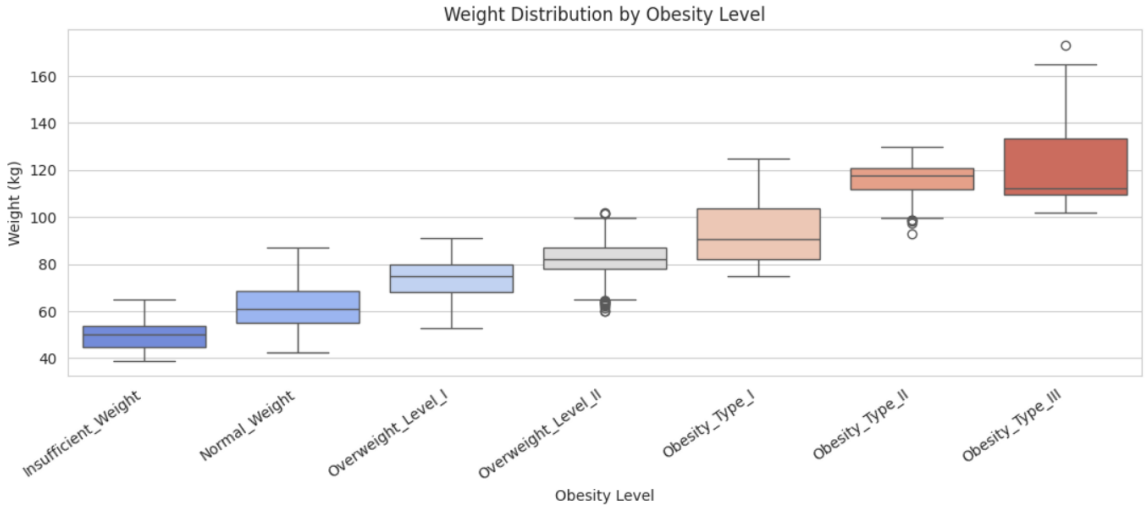

Feature Correlation with Obesity Level
Weight
family_history_with_overweight
Age
FAVC
FCVC
CALC
Height
CH2O
MTRANS_Public_Transportation
NCP
SMOKE
MTRANS_Automobile
Gender
MTRANS_Bike
MTRANS_Motorbike
TUE
MTRANS_Walking
SCC
FAF
CAEC
−0.2
0.0
0.2
0.4
0.6
0.8


Distribution of Obesity Levels
Count
0
50
100
150
200
250
300
350
272
287
290
290
351
297
324
Insufficient_Weight
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III
Obesity Level

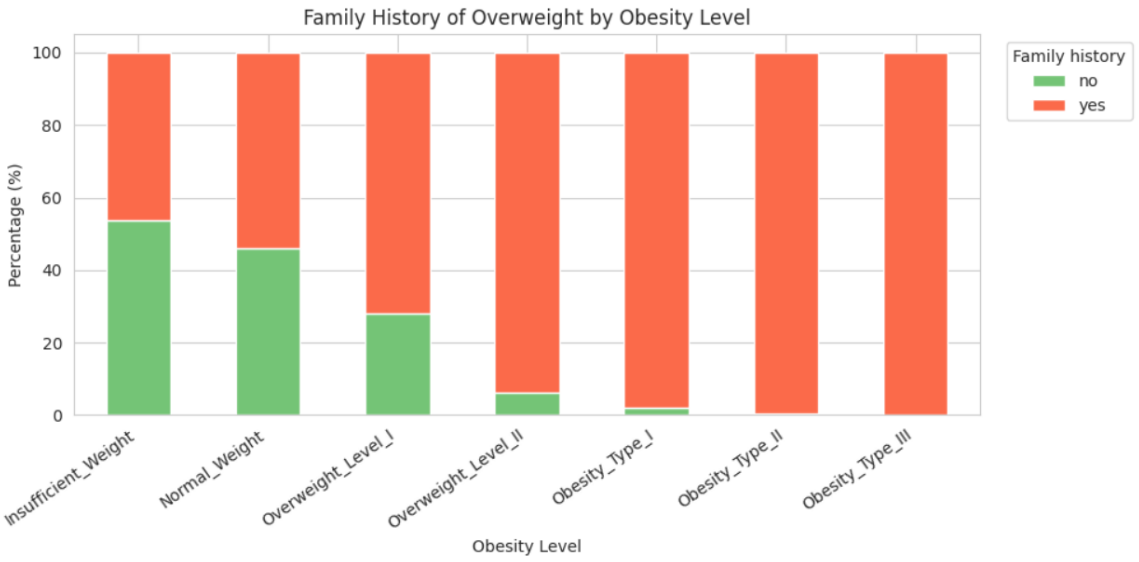

Obesity Levels by Gender
Gender
Female
Male
Count
0
50
100
150
200
250
300
Insufficient_Weight
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III
Obesity Level

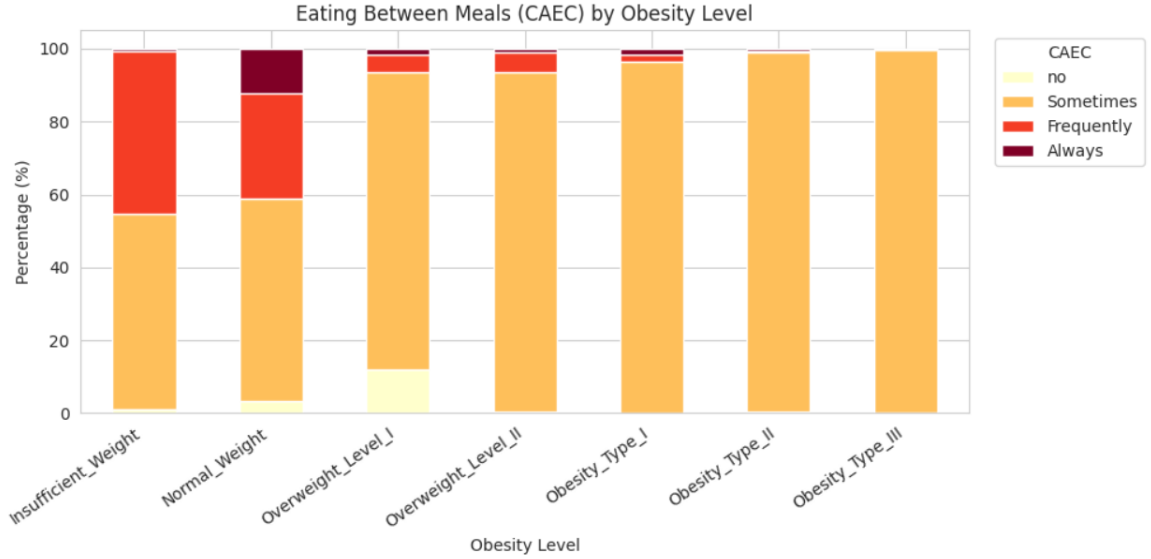

Age Distribution
Frequency
0
50
100
150
200
250
300
20
30
40
50
60
Age

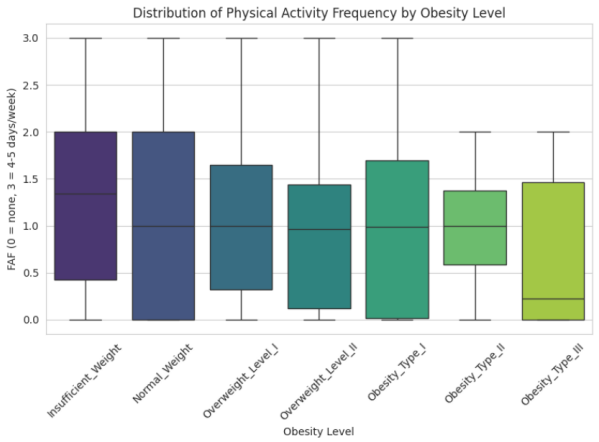

Height vs Weight by Obesity Level
Weight
160
140
120
100
80
60
40
Insufficient_Weight
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III

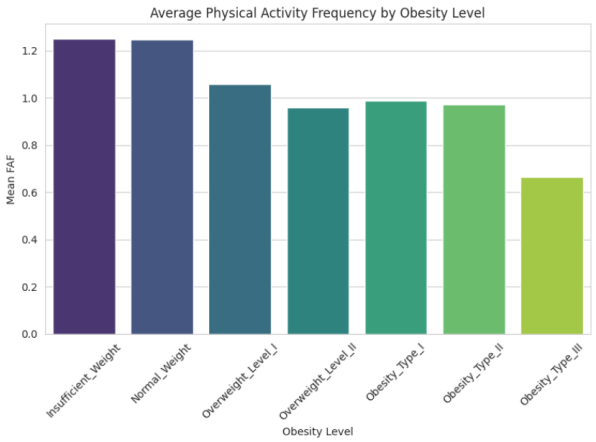

Weight Distribution by Obesity Level
Weight (kg)
160
140
120
100
80
60
40
Insufficient_Weight
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III
Obesity Level

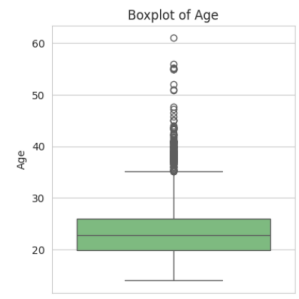
...

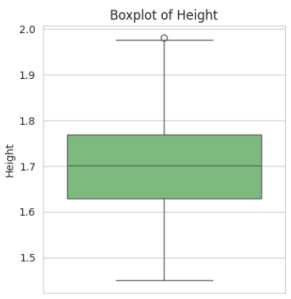

Family History of Overweight by Obesity Level
Percentage (%)
0
20
40
60
80
100
Insufficient_Weight
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III
Obesity Level
Family history
no
yes

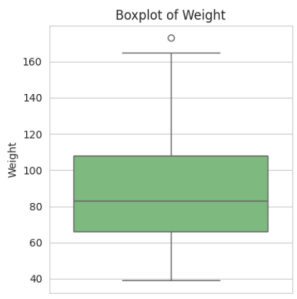

Eating Between Meals (CAEC) by Obesity Level
Percentage (%)
0
20
40
60
80
100
Insufficient_Weight
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III
Obesity Level
CAEC
no
Sometimes
Frequently
Always

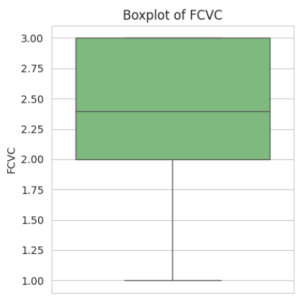

Distribution of Physical Activity Frequency by Obesity Level
FAF (0 = none, 3 = 4-5 days/week)
Insufficient_Weight
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III
Obesity Level

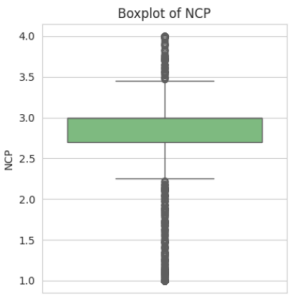

Average Physical Activity Frequency by Obesity Level
Mean FAF
Insufficient_Weight
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III
Obesity Level

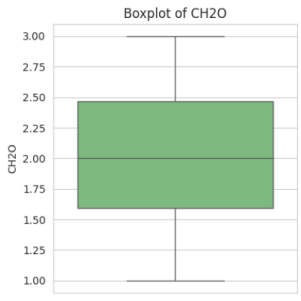

Boxplot of Age
Age

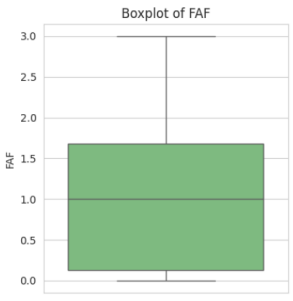

Boxplot of Height
Height

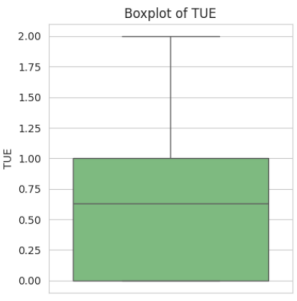

Boxplot of Weight
Weight

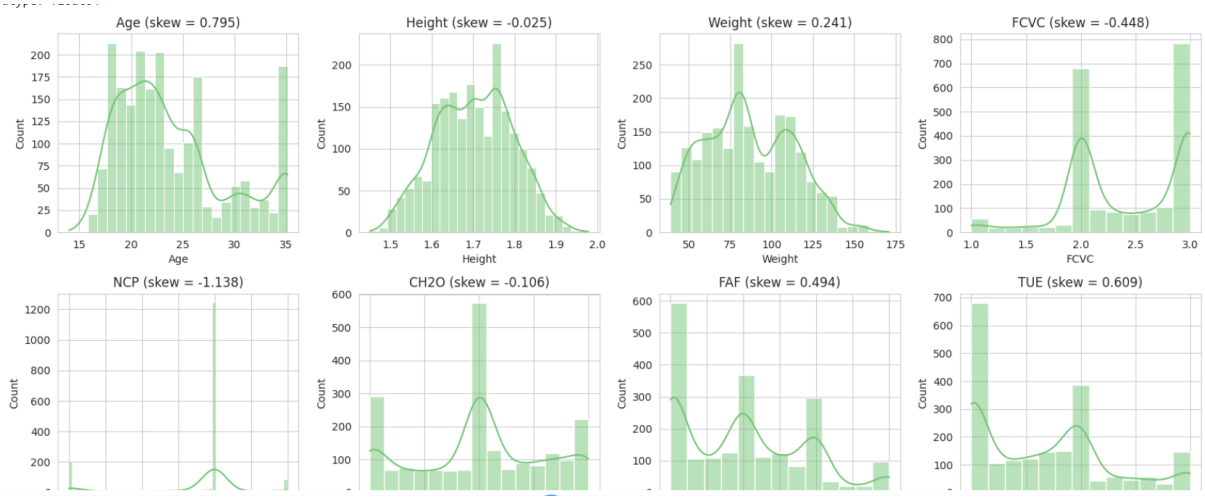

Boxplot of FCVC
FCVC

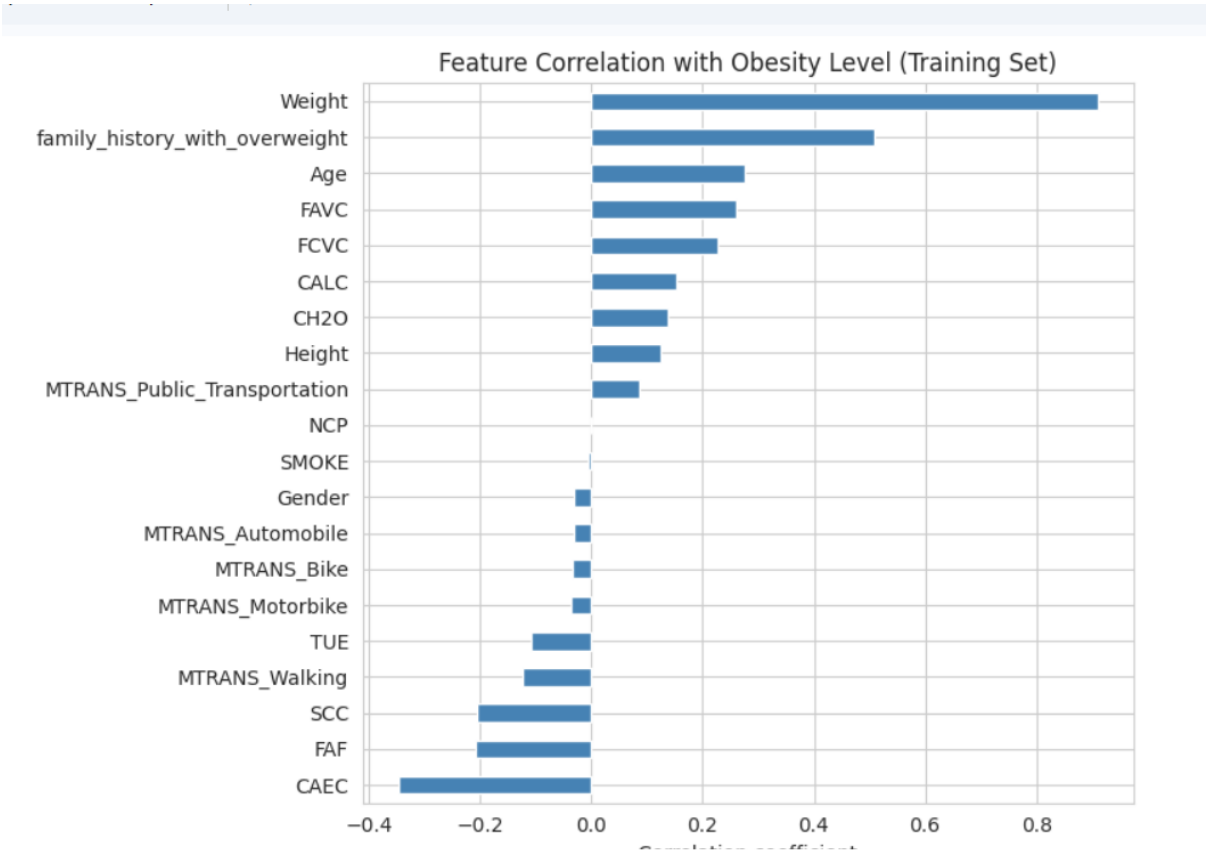

Boxplot of NCP
NCP


Boxplot of CH2O
CH2O

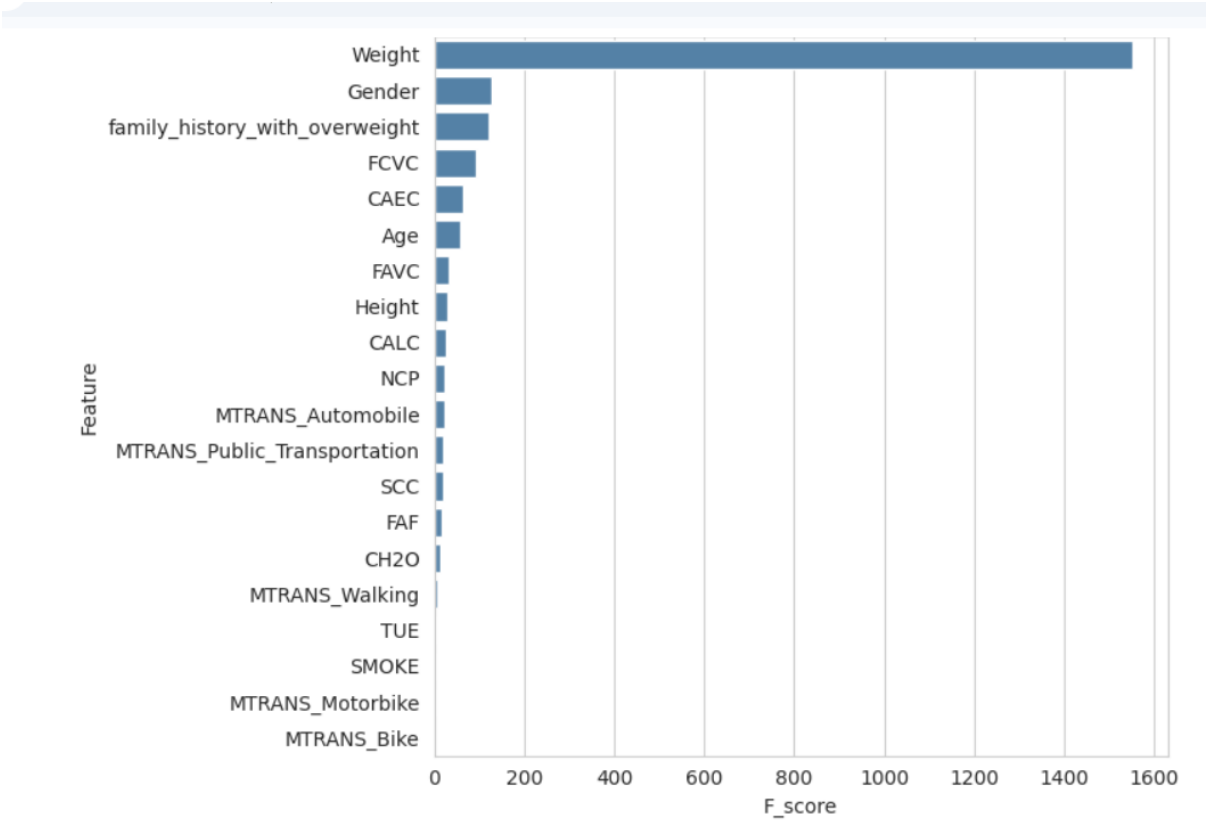

Boxplot of FAF
FAF

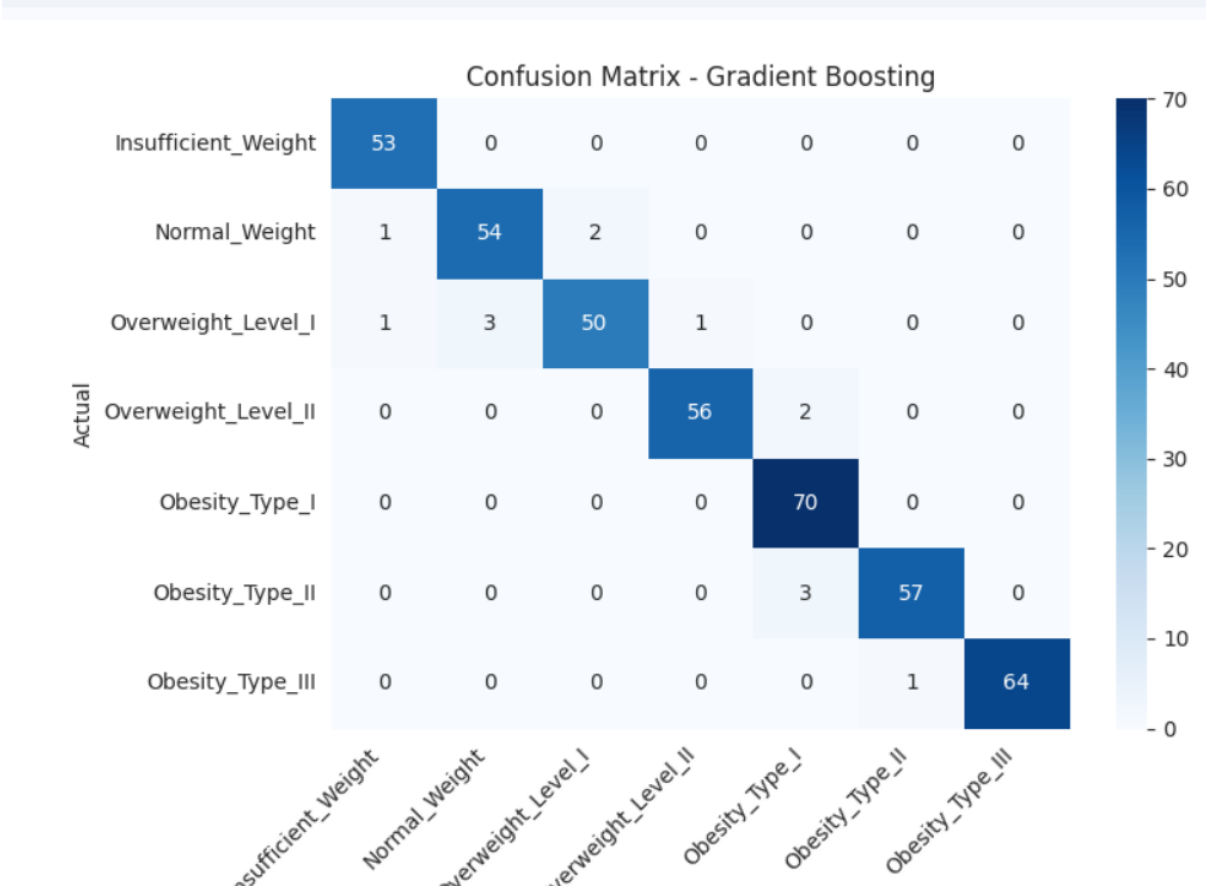

Boxplot of TUE
TUE

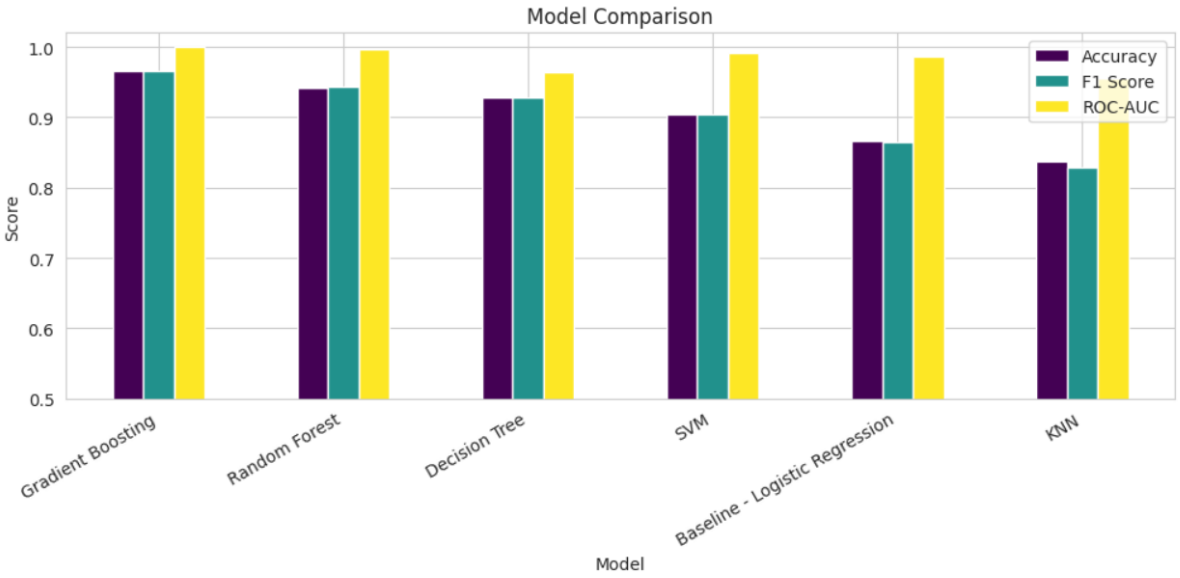


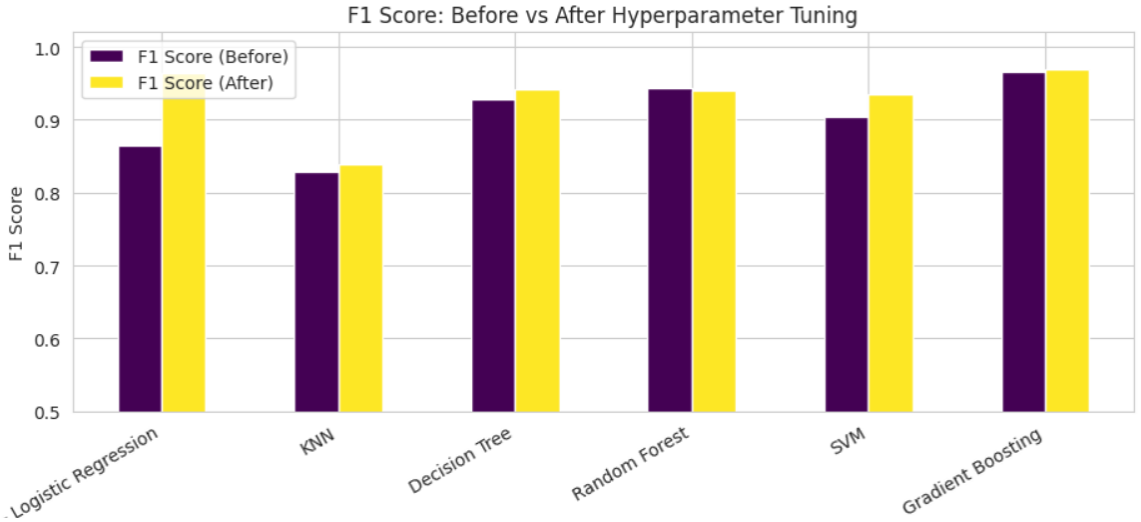


Select K best by features


Weight
Gender
family_history_with_overweight
FCVC
CAEC
Age
FAVC
Height
CALC
NCP
MTRANS_Automobile
MTRANS_Public_Transportation
SCC
FAF
CH2O
MTRANS_Walking
TUE
SMOKE
MTRANS_Motorbike
MTRANS_Bike
Feature
0
200
400
600
800
1000
1200
1400
1600
F_score


Confusion Matrix - Gradient Boosting
Insufficient_Weight
Normal_Weight
Overweight_Level_I
Overweight_Level_II
Obesity_Type_I
Obesity_Type_II
Obesity_Type_III
Actual
53 0 0 0 0 0 0
1 54 2 0 0 0 0
1 3 50 1 0 0 0
0 0 0 56 2 0 0
0 0 0 0 70 0 0
0 0 0 0 3 57 0
0 0 0 0 0 1 64
70
60
50
40
30
20
10
0


Model Comparison
Accuracy
F1 Score
ROC-AUC
Score
1.0
0.9
0.8
0.7
0.6
0.5
Gradient Boosting
Random Forest
Decision Tree
SVM
Baseline - Logistic Regression
KNN
Model


F1 Score: Before vs After Hyperparameter Tuning
F1 Score (Before)
F1 Score (After)
F1 Score
1.0
0.9
0.8
0.7
0.6
0.5
Logistic Regression
KNN
Decision Tree
Random Forest
SVM
Gradient Boosting